Recipe Site Traffic

Predicting which recipes are likely to generate high traffic

Recipe Site | High traffic recipes

Business Goal

Turn recipe traffic data into better recipe recommendations

Identify high-traffic recipes

Predict which recipes are likely to attract high traffic before they are featured.

Meet the 80% requirement

Correctly identify at least 80% of recipes that generate high traffic.

Avoid poor recommendations

Minimize low-traffic recipes being incorrectly selected as High traffic.

Modeling question

Can recipe characteristics be used to predict whether a recipe will generate High or Low traffic?

Data & Validation

947 recipes and six predictive variables

Recipes	Predictors	Traffic classes	Missing nutrition
947	6	2	52

Key data issues

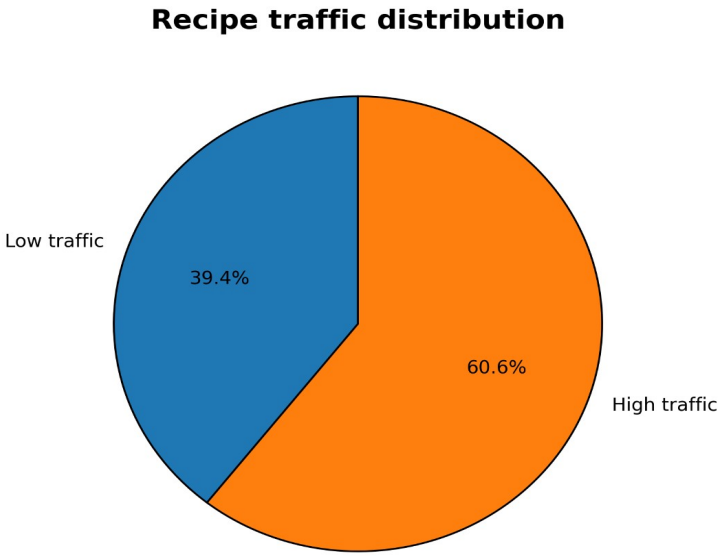
- 11 categories instead of the expected 10
- Servings stored as text such as “4 as a snack”
- 52 recipes missing the four nutrition variables
- High-traffic field contained High / missing values

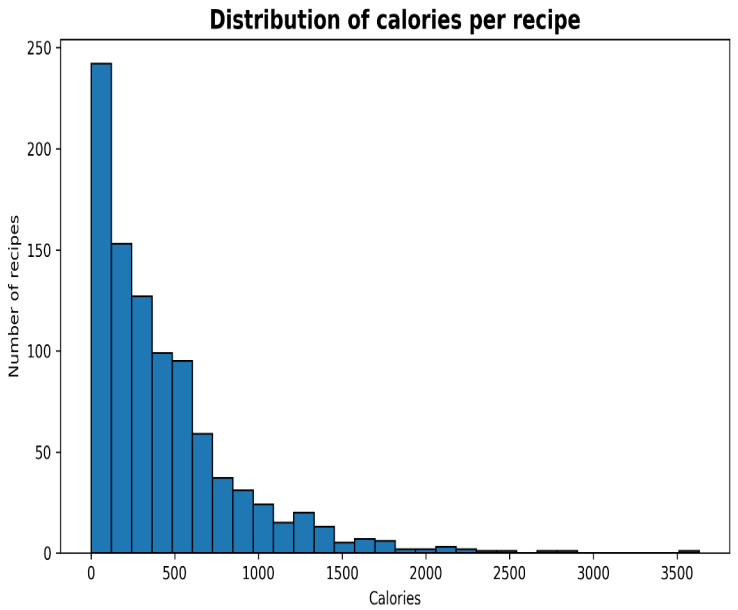
Cleaning decisions

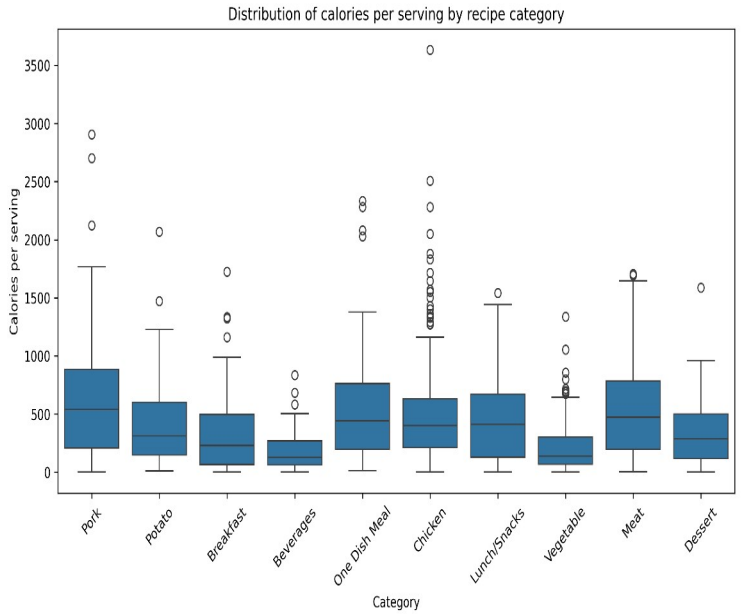
- Merged Chicken Breast → Chicken
- Extracted numeric serving size
- Imputed nutrition using category × servings group means
- Converted traffic to binary High = 1 / Low = 0

What did the data show?

Key exploratory findings







Traffic

About 60% of recipes were classified as High traffic.

Calories

Calories are strongly right-skewed with a long upper tail.

Categories

Calorie distributions vary across recipe categories.

Model Development

Two approaches were fitted using the same data split

1. Logistic Regression

Baseline model for binary classification

Simple and interpretable

Provides a clear benchmark for the problem

2. XGBoost

Comparison model using tree-based boosting

Can capture non-linear relationships and interactions

Constrained to reduce the risk of overfitting

Evaluation process

80% training → 10% validation → 10% test. Models were fitted on training data, the classification threshold was optimized on validation data, and the untouched test set was used for final evaluation.

Model Evaluation

Final performance on the unseen test set

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	74.7%	74.6%	87.7%	80.6%
XGBoost	70.5%	73.0%	80.7%	76.7%

Logistic Regression

Slightly higher performance across all four test metrics.

XGBoost

Comparable overall performance, but lower test precision and recall.

Business Metrics

Translate model performance into the business decision

Recall ≥ 80%

Minimum requirement: identify enough genuinely high-traffic recipes.

Precision as high as possible

Primary KPI for reducing low-traffic recipes incorrectly recommended as High.

The trade-off

Higher recall can require more High predictions, increasing false positives and reducing precision.

Model	Precision	Recall
Logistic Regression	74.6%	87.7%
XGBoost	73.0%	80.7%

Both models met the 80% recall requirement; Logistic Regression also achieved slightly higher precision.

Recommendation

A simpler model with slightly better test performance

Selected model

Logistic Regression

Test precision

74.6%

Test recall

87.7%

Use as the initial solution

- Slightly higher test performance than XGBoost
- Simpler and easier to interpret
- No clear predictive advantage from the more complex model

Monitor & refine

- Monitor precision and recall after deployment
- Retrain as new traffic data becomes available
- Review the 80% recall constraint if false positives prove especially costly

Final Takeaways

1. Data

The data was cleaned and validated.

Key inconsistencies and missing values were addressed.

EDA revealed differences across recipe categories.

2. Models

Both models satisfied the minimum recall requirement.

Logistic Regression achieved slightly better test metrics.

Its simpler structure makes it a practical initial choice.

3. Next steps

Monitor precision and recall.

Collect more real traffic outcomes.

Use observed business costs to refine the threshold.

Thank you